



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

SCORE - II

25-05-2021

JEE(Main + Advance) : Enthusiast Course (PHASE : SCORE-II)

Test Type : SCORE - II

Test Pattern : JEE (Advanced)

TEST DATE : 25-05-2021

ANSWER KEY

PAPER-2

PHYSICS

SEC-1	Q.	1	2	3	4	5			
	A.	C	A	D	D	C			
SEC-2	Q.	1	2	3	4	5	6	7	8
	A.	A,C,D	A,B,C,D	A,C	A,C,D	A,B,C,D	B,C,D	A,D	A,B,C
SEC-3	Q.	1	2	3	4	5			
	A.	331	231	211	213	111			

CHEMISTRY

SEC-1	Q.	1	2	3	4	5			
	A.	D	C	A	B	A			
SEC-2	Q.	1	2	3	4	5	6	7	8
	A.	A,C	A	A,D	A,B,C,D	B,D	B,C,D	B,C,D	A,C
SEC-3	Q.	1	2	3	4	5			
	A.	231	312	233	312	213			

MATHEMATICS

SEC-1	Q.	1	2	3	4	5			
	A.	C	D	B	A	B			
SEC-2	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C	B,C	D	A,D	A,B,D	A,C	B,C,D	B,C,D
SEC-3	Q.	1	2	3	4	5			
	A.	111	123	132	123	132			

HINT – SHEET

PHYSICS

SEC-1

1. Ans (C)

Refer HCV part-I, page no. 244

2. Ans (A)

$$V_{P \max} = A\omega \sin\left(\frac{\pi}{6}\right)$$

$$\omega = 2\pi f = 2\pi \frac{v}{\lambda} \text{ \& } \lambda = 0.4 \text{ m}$$

3. Ans (D)

$$I = I_0 + I_0 + 2\sqrt{I_0 I_0} \cos \phi,$$

$$\phi = \frac{2\pi}{\lambda} \Delta x = \frac{2\pi}{\lambda} \frac{\lambda}{4} = \frac{\pi}{2}$$

$$\text{so } I = 2I_0$$

4. Ans (D)

$$d \sin \theta = n\lambda$$

$$d = \frac{n\lambda}{\sin \theta} = \frac{\lambda}{\sin \theta} = \frac{0.12}{3} \times 5$$

$$d = \frac{0.6}{3} = 0.2 \text{ m}$$

5. Ans (C)

Lenses are of opposite nature, so,

$$\text{in water, } \frac{1}{f_1} = \left(\frac{\mu_1}{\mu_w} - 1 \right) \left(\frac{2}{R} \right)$$

$$\frac{1}{f_2} = \left(\frac{\mu_2}{\mu_w} - 1 \right) \left(-\frac{2}{R} \right)$$

focal length of the system is

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\frac{1}{24} = \left(\frac{\mu_1}{\mu_w} - \frac{\mu_2}{\mu_w} \right) \left(\frac{2}{R} \right)$$

$$= \left(\frac{\mu_1 - \mu_2}{\mu_w} \right) \left(\frac{2}{20} \right)$$

$$\text{So, } (\mu_1 - \mu_2) = \frac{5}{9}$$

PHYSICS

SEC-2

1. Ans (A,C,D)

$$T_A = T_B = (300 + 400)/2 = 350^\circ\text{C}$$

$$\therefore T_A = T_B = T_C$$

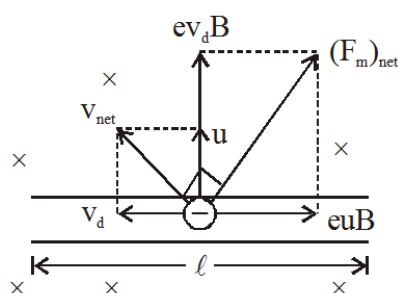
2. Ans (A,B,C,D)

$$Ft = mv$$

$$W = \frac{1}{2}mv^2 = \frac{1}{2} \frac{mF^2 t^2}{m^2}$$

$$W = \frac{F^2 t^2}{2m}$$

6. Ans (B,C,D)



$$W_{\text{magnetic}} = 0$$

To maintain the current work has to be done by battery

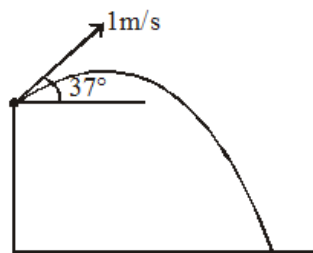
$$W_{\text{batt}} = \int (nA\ell)euBv_d dt$$

$$I = neAv_d$$

$$= I\ell B \int u dt$$

$$W_{\text{Batt}} = I\ell Bh$$

7. Ans (A,D)



$$v = \sqrt{2gh} = \sqrt{2 \times 10 \times \frac{5}{100}} = 1 \text{ m/s}$$

$$\frac{-8}{100} = \frac{3}{5}t - \frac{1}{2} \times 10 \times t^2$$

$$\Rightarrow t = \frac{1}{5} \text{ sec, } \frac{-8}{100} \text{ sec}$$

$$\therefore x = \frac{4}{5} \times \frac{1}{5} \times 100 = 16 \text{ cm}$$

$$\therefore \text{horizontal range from point B} = 16 - 6 = 10 \text{ cm}$$

Force required to keep the vessel in static equilibrium.

$$= \rho A v^2 \cos 37^\circ$$

$$= 1000 \times \frac{3}{2} \times 10^{-4} \times 1 \times \frac{4}{5} = 0.12 \text{ N}$$

8. Ans (A,B,C)

$$d = 2a(\mu - 1)\alpha$$

$$\beta = \frac{D\lambda}{d} \Rightarrow \frac{D\lambda}{2a(\mu - 1)\alpha} = \beta$$

(A) α is decreased so $\beta \uparrow$

$$(B) \beta' = \frac{D(\lambda/\mu_m)}{2a\alpha \left(\frac{\mu}{\mu_m} - 1 \right)} = \frac{D\lambda}{2a\alpha} \frac{1}{(\mu - \mu_m)}$$

$$\text{Now, } (\mu - \mu_m) < (\mu - 1)$$

so, $\beta \uparrow$

(C) $\alpha \downarrow \Rightarrow \beta \uparrow$

PHYSICS

SEC-3

1. Ans (331)

(A) For system A + B, $(4 + 6) v_0 = 4 \times 5 + 6 \times 10$

$$\Rightarrow v_0 = 8 \text{ m/s}$$

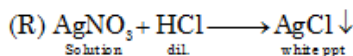
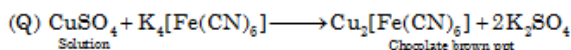
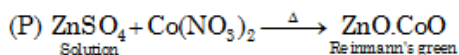
(B) $v_{\text{CM}} =$

$$\frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} = \frac{4 \times 5 + 6 \times 10}{4 + 6} = 8 \text{ m/s}$$

(C) Work done by friction on

$$B = \frac{1}{2} \times 6 \times 8^2 - \frac{1}{2} \times 6 \times 10^2 = -108 \text{ J}$$

4. Ans (312)



MATHEMATICS

SEC-1

1. Ans (C)

Given $f_{(n+1)}(x) = e^{f_n(x)}$

$\Rightarrow f'_{n+1}(x) = e^{f_n(x)} f'_n(x)$

$\Rightarrow f'_{(n+1)}(x) = f_{(n+1)}(x) f'_n(x)$

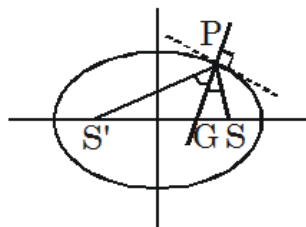
$\Rightarrow f'_n(x) = f_n(x) f'_{(n-1)}(x)$

$= f_n(x) f_{(n-1)}(x) \dots f_2(x) f'_1(x)$

$= f_n(x) f_{(n-1)}(x) \dots f_1(x)$

2. Ans (D)

Since PG is the internal angle bisector of $\angle \text{SPS}'$



$\Rightarrow \frac{SG}{S'G} = \frac{SP}{S'P} = \frac{2}{8} = \frac{1}{4}$

$\Rightarrow \frac{\text{ar}(\triangle \text{PSG})}{\text{ar}(\triangle \text{PS}'G)} = \frac{1}{4}$

3. Ans (B)

Let $P_0 = 130$ crore

After n years, population $P = P_0(1 + 0.01)^n$

Total number of possible mobile numbers $= 4 \times 10^9 = 400$ crore

Hence, $2P \leq 400$ crore

$\Rightarrow 130(1.01)^n \leq 200 \Rightarrow n \leq \log_{1.01} \left(\frac{20}{13} \right) = 43.29$

$\Rightarrow n \leq 43$ years

4. Ans (A)

Out of all 4C_2 combinations,

only combination $\sin \frac{1}{x} \cos \frac{1}{x}$ does not has limit at $x = 0$.

So, $P(E) = \frac{{}^4C_2 - 1}{{}^4C_2} = \frac{5}{6}$

5. Ans (B)

Let $z = e^{i\theta}$

So, $w = \frac{e^{i\theta} - 1}{e^{i\theta} + 1} = \frac{(e^{i\theta} - 1)(e^{-i\theta} + 1)}{|e^{i\theta} + 1|^2}$

$= \frac{e^{i\theta} - e^{-i\theta}}{2 + 2\cos\theta} = \frac{2i\sin\theta}{2(1 + \cos\theta)} = i \tan\left(\frac{\theta}{2}\right)$

MATHEMATICS

SEC-2

1. Ans (A,B,C)

$(\sin^{-1}x)^2 - (\cos^{-1}x)^2 = \frac{\pi}{2}(\sin^{-1}x - \cos^{-1}x)$

$= \frac{\pi}{2} \left(\frac{\pi}{2} - 2\cos^{-1}x \right) \in \left[-\frac{3\pi^2}{4}, \frac{\pi^2}{4} \right] \subset (-8, 3)$

Also, $\sin^{-1}x + 2\cos^{-1}(-x) = 2\pi$

$\Rightarrow \sin^{-1}x + 2(\pi - \cos^{-1}x) = 2\pi$

$\Rightarrow \sin^{-1}x - 2 \left(\frac{\pi}{2} - \sin^{-1}x \right) = 0$

$\Rightarrow \sin^{-1}x = \frac{\pi}{3} \Rightarrow x = \frac{\sqrt{3}}{2} > \frac{1}{2}$

2. Ans (B,C)

$A^2 = I \Rightarrow (\det A)^2 = \det(I)$

$\Rightarrow \det A = \pm 1$

so, $\det((A - I)^3 + (A + I)^3 - 7A)$

$= \det(2A^3 + 6A - 7A) = \det(2A^2A - A)$

$= \det(A) = \pm 1$

3. Ans (D)

Function $|t|$ is not differentiable only at $t = 0$,

whereas function $|\sin \pi t|$ is differentiable

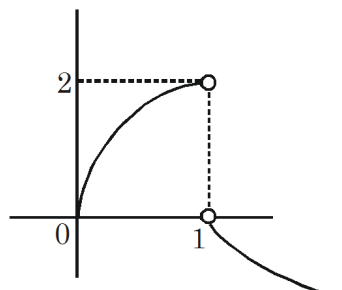
everywhere except at all integral points. Also,

function $\sqrt[3]{t-2}$ is not differentiable only at $t = 2$.

So, $\vec{r}(t)$ is differentiable everywhere except at all integral points.

4. Ans (A,D)

At $x = 1$, $f(x)$ has local
 $\begin{cases} \text{maxima ; if } p \geq 2 \\ \text{minima ; if } p < 0 \end{cases}$



5. Ans (A,B,D)

$$\int e^x (x+1)^2 dx = \int e^x ((x^2+1) + 2x) dx$$

$$= e^x(x^2+1) + c$$

$$(\text{as } \int e^x (f + f') dx = e^x f + c)$$

6. Ans (A,C)

Given identity

$$\int_0^\infty \frac{x^2}{(x^2+a^2)(x^2+b^2)(x^2+c^2)} dx$$

$$= \frac{\pi}{2(a+b)(b+c)(c+a)}$$

For I, put $a=0, b=2$ and $c=3$:

$$\int_0^\infty \frac{dx}{(x^2+4)(x^2+9)} = \frac{\pi}{60} = I \quad \dots(i)$$

Now, put $a=1, b=2$ and $c=3$:

$$\int_0^\infty \frac{x^2 dx}{(x^2+1)(x^2+4)(x^2+9)} = \frac{\pi}{120} \quad \dots(ii)$$

so, (i)-(ii):

$$\int_0^\infty \frac{dx}{(x^2+1)(x^2+4)(x^2+9)} = \frac{\pi}{120} = J$$

7. Ans (B,C,D)

$$\Delta = \begin{vmatrix} \sin \theta & \cos \theta & \cos \theta \\ \cos \theta & \sin \theta & \cos \theta \\ \cos \theta & \cos \theta & \sin \theta \end{vmatrix}$$

$$= (\sin \theta + 2 \cos \theta) \begin{vmatrix} 1 & 1 & 1 \\ \cos \theta & \sin \theta & \cos \theta \\ \cos \theta & \cos \theta & \sin \theta \end{vmatrix}$$

$$[R_1 \rightarrow R_1 + R_2 + R_3]$$

=

$$(\sin \theta + 2 \cos \theta) \begin{vmatrix} 1 & 0 & 0 \\ \cos \theta & \sin \theta - \cos \theta & 0 \\ \cos \theta & 0 & \sin \theta - \cos \theta \end{vmatrix}$$

$$\begin{bmatrix} C_2 \rightarrow C_2 - C_1 \\ C_3 \rightarrow C_3 - C_1 \end{bmatrix}$$

$$= (\sin \theta - \cos \theta)^2 (\sin \theta + 2 \cos \theta)$$

Now, if $\tan \theta = -2$ then $\Delta = 0$

$$\text{But, } h^2 - ab = \cos^2 \theta - \sin^2 \theta$$

$$= \cos 2\theta = -\frac{3}{5} < 0$$

i.e. imaginary pair of lines

Hence, for real straight lines, $\theta = n\pi + \frac{\pi}{4}$

Also, in this case $h^2 - ab = \cos 2\theta = 0$

i.e. lines are parallel.

further, one can observe that by taking $\theta = \frac{\pi}{2}$,

$$S = 0 \text{ becomes } x^2 + y^2 + 1 = 0$$

i.e. no real locus.

8. Ans (B,C,D)

Tangent at any point (x,y) is

$$Y - y = y'(X - x)$$

$$\text{so, } -y = y'(1 - x) \Rightarrow \frac{dy}{dx} = \frac{y}{x-1}$$

$$\text{It leads to } \int \frac{dy}{y} = \int \frac{dx}{x-1}$$

$$\Rightarrow y = m(x-1)$$

MATHEMATICS

SEC-3

1. **Ans (111)**

$$(P) \quad \frac{x + x + y}{3} \geq (x^2y)^{1/3} \Rightarrow x^2y \leq 1$$

$$(Q) \quad \sin 0^\circ < \sin 25^\circ < \sin 30^\circ$$
$$\Rightarrow -1 < -2\sin 25^\circ < 0 \Rightarrow [-2 \sin 25^\circ] = -1$$

(R)

$$\begin{aligned} \left| \frac{\sqrt{2} + \sqrt{3}w + \sqrt{5}w^2}{\sqrt{5} + \sqrt{2}w + \sqrt{3}w^2} \right| &= |w| \left| \frac{\sqrt{2} + \sqrt{3}w + \sqrt{5}w^2}{\sqrt{5} + \sqrt{2}w + \sqrt{3}w^2} \right| \\ &= \left| \frac{\sqrt{2}w + \sqrt{3}w^2 + \sqrt{5}}{\sqrt{5} + \sqrt{2}w + \sqrt{3}w^2} \right| = 1 \end{aligned}$$

2. Ans (123)

(P) $\vec{r}$ cannot be null vector.

Also, $\vec{r} \times \hat{i}$ is perpendicular to $\hat{i}$.

Hence $\vec{r} \times \hat{i} \neq \hat{i}$

(Q) $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 1 \Rightarrow x + y + z = 1$ (A plane) so, length of perpendicular distance of origin from this plane $= |\vec{r}| = \frac{1}{\sqrt{3}}$

Hence, there is only one point i.e. foot of normal drawn from the origin to the plane, satisfying the given equation.

(R) There are infinite points in the space which lies at unit distance from $(1,0,0)$

3. Ans (132)

(P) $\sin^2 x - 5 \sin x + 6 < 0$

$$\Rightarrow \sin x \in [2,3]; \text{ which is not possible}$$

$$(Q) \quad x^2 - \frac{y^2}{24} = 1 \Rightarrow e = \sqrt{24 + 1} = 5$$

(R) Let $y = mx + \frac{1}{m}$ be any tangent

$$\text{so, } 4 = m + \frac{1}{m} \Rightarrow m^2 - 4m + 1 = 0$$

$$\Rightarrow |\tan \theta| = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \frac{\sqrt{16-4}}{2} = \sqrt{3}$$

$$\Rightarrow (\text{as } \theta \text{ is obtuse})$$

Hence, $\theta = \frac{2\pi}{3}$

4. Ans (123)

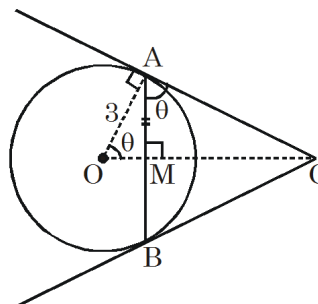
(P) Required plane is

$$\begin{vmatrix} x+3 & y-1 & z+2 \\ 1 & 2 & 3 \\ 1 & -2 & 3 \end{vmatrix} = 0$$

$$\Rightarrow 12(x + 3) - 4(z + 2) = 0$$

$$\Rightarrow 3x - z + 7 = 0$$

$$(Q) OM = \sqrt{5} \Rightarrow AM = 2$$



$$\text{So, CM} = \text{AM} \tan \theta = 2 \times \frac{2}{\sqrt{5}}$$

$$\Rightarrow \text{ar}(\triangle ABC) = \frac{1}{2} \times AB \times CM$$
$$= \frac{1}{2} \times 4 \times \frac{4}{\sqrt{5}} = \frac{8}{\sqrt{5}}$$

$$(R) \quad N = \frac{|5|}{|2|} = 60$$

5. Ans (132)

(P) Given sum = coefficient of x^5 in the expansion of

$$(1 - x)^{10}(1 + x + x^2)^{10} \text{ Which is } (1 - x^3)^{10}$$

So, required sum = 0 (as there is no any term containing x^5)

(Q) $f(x) = e^{x-[x]} = e^{\{x\}}$, which is periodic with period 1

$$(R) \quad \frac{dy}{dx} = -3x^2 + 6x + 2 = 5 - 3(x-1)^2 \leq 5$$